22GE004-BASICS OF ELECTRONICS ENGINEERING

UNIT II & LP8-SIGNAL CONDITIONING USING DIODE

1. LIMITATIONS OF PN JUNCTION DIODE

- If a diode is being used as a power rectifier and the reverse voltage is high enough to cause it to avalanche, it is usually a disaster for the diode.
- The reverse current is suddenly not restricted and the diode self-destructs.
- When used normally as a zener, this current is controlled by the series resistor.
- Semiconductor diode cannot withstand very high reverse voltage.
- It offers poor response.
- It has reverse saturation current.
- Noise level is high in semiconductor device at high frequencies.
- Power loss: Although diodes are efficient in terms of power consumption, there is still a power loss associated with them.
- This loss is due to the voltage drop across the diode, which can generate heat and affect the overall performance of the circuit.
- Frequency limitations: Diodes have limitations in terms of operating frequency.